

# CS 2601 Linear and Convex Optimization

## 13. Barrier method

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# General smooth convex problems

Consider

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b} \\ & g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, m\end{array} \quad (\text{ICP})$$

where  $f, g_j$  are smooth convex functions.

So far, we can solve some special cases

- When projection onto the feasible set is easy, we can use projected gradient descent
- When there are no inequality constraints, we can use Newton's method or projected gradient descent

For the general (ICP), we are going to get rid of the inequality constraints, but account for their effects by modifying the objective function.

## Reformulation

Let  $\bar{S}$  be the set defined by the inequality constraints,

$$\bar{S} = \{\mathbf{x} : g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, m\}$$

(ICP) is equivalent to

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}) + I_{\bar{S}}(\mathbf{x}) \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b} \end{array}$$

where  $I_{\bar{S}}(\mathbf{x})$  is the indicator of  $\bar{S}$

$$I_{\bar{S}} = \begin{cases} 0, & \mathbf{x} \in \bar{S} \\ +\infty, & \mathbf{x} \notin \bar{S} \end{cases}$$

We will approximate  $I_{\bar{S}}(\mathbf{x})$  by so-called **barrier functions**.

# Barrier function

Let  $S$  be the interior of  $\bar{S}$ ,

$$S = \{\mathbf{x} : g_j(\mathbf{x}) < 0, j = 1, 2, \dots, m\}$$

Assume it is nonempty. A continuous function  $B : S \rightarrow \mathbb{R}$  is a **barrier function** for  $S$  if  $B(\mathbf{x}) \rightarrow +\infty$  as  $\mathbf{x} \in S$  approaches the boundary of  $S$

Common barrier functions

- logarithmic barrier function

$$B(\mathbf{x}) = - \sum_{j=1}^m \log(-g_j(\mathbf{x}))$$

- inverse barrier function

$$B(\mathbf{x}) = - \sum_{j=1}^m \frac{1}{g_j(\mathbf{x})}$$

Note both functions are convex. We will use the log barrier function.

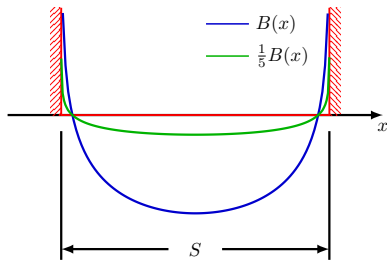
# Log barrier function

Example.  $S = (a, b)$  with

$$g_1(x) = a - x < 0$$

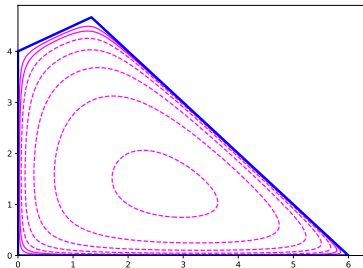
$$g_2(x) = x - b < 0$$

$$B(x) = -\log(x - a) - \log(b - x)$$



Example.  $S = \{\mathbf{x} : \mathbf{Ax} < \mathbf{b}\}$ , where

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -1 & 2 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 6 \\ 8 \\ 0 \\ 0 \end{bmatrix}$$



$$B(\mathbf{x}) = -\log(6 - x_1 - x_2) - \log(8 + x_1 - 2x_2) - \log(x_1) - \log(x_2)$$

## Log barrier function (cont'd)

$$B(\mathbf{x}) = -\sum_{j=1}^m \log(-g_j(\mathbf{x}))$$

gradient

$$\nabla B(\mathbf{x}) = \sum_{j=1}^m \frac{1}{-g_j(\mathbf{x})} \nabla g_j(\mathbf{x})$$

Hessian

$$\nabla^2 B(\mathbf{x}) = \sum_{j=1}^m \frac{1}{g_j^2(\mathbf{x})} \nabla g_j(\mathbf{x}) \nabla g_j(\mathbf{x})^T + \sum_{j=1}^m \frac{1}{-g_j(\mathbf{x})} \nabla^2 g_j(\mathbf{x})$$

**Proof.** Suffices to prove for  $m = 1$  i.e.  $B(\mathbf{x}) = -\log(-g(\mathbf{x}))$ . Use

$$\partial_\ell \left( \frac{\partial_k g}{-g} \right) = \frac{\partial_\ell g \partial_k g}{g^2} + \frac{\partial_\ell \partial_k g}{-g} = \frac{[\nabla g \nabla g^T]_{k\ell}}{g^2} + \frac{[\nabla^2 g]_{k\ell}}{-g}$$

## Approximation

We approximate  $I_{\bar{S}}(\mathbf{x})$  by  $\frac{1}{t}B(\mathbf{x})$  and (ICP) by

$$\begin{array}{ll} \min_{\mathbf{x}} & F(\mathbf{x}) = f(\mathbf{x}) + \frac{1}{t}B(\mathbf{x}) \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b} \end{array} \quad (\text{B})$$

where the parameter  $t > 0$  controls the approximation accuracy.

**Note.** For (B) to be feasible, (ICP) should be **strictly feasible** i.e.  $\{\mathbf{x} \in D : \mathbf{Ax} = \mathbf{b}, g_j(\mathbf{x}) < 0, j = 1, \dots, m\}$  is nonempty. We assume so.

We can solve (B) by e.g. damped Newton's method.

**Note.**  $\text{dom } F \neq \mathbb{R}^n$  (why?). Always check whether  $\mathbf{x}_k + t\mathbf{d}_k \in \text{dom } F$  during backtracking line search. This can also be taken care of by defining  $F(\mathbf{x}) = +\infty$  for  $\mathbf{x} \notin \text{dom } F$ .

## Example: LP

$$\begin{array}{ll}\min_{\mathbf{x} \in \mathbb{R}^2} & f(\mathbf{x}) = -x_1 - 3x_2 \\ \text{s.t.} & x_1 + x_2 \leq 6 \\ & -x_1 + 2x_2 \leq 8 \\ & x_1, x_2 \geq 0\end{array}$$

The approximation is an unconstrained problem

$$\min F(\mathbf{x})$$

with

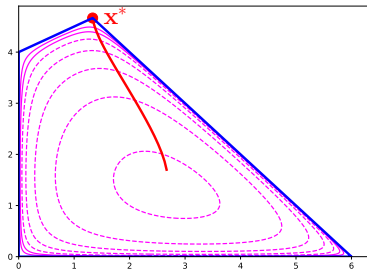
$$F(\mathbf{x}) = -x_1 - 3x_2 + \frac{1}{t} [-\log(6 - x_1 - x_2) - \log(8 + x_1 - 2x_2) - \log(x_1) - \log(x_2)]$$

**Note.** We actually have implicit constraints.

$$\text{dom } F = S = \{\mathbf{x} : x_1 + x_2 < 6, -x_1 + 2x_2 < 8, x_1 < 0, x_2 < 0\} \neq \mathbb{R}^2$$

## Central path

The solution  $\mathbf{x}^*(t)$  to (B) is called a **central point**. As  $t > 0$  varies, the curve  $\mathbf{x}^*(t)$  is called the **central path** of (ICP).

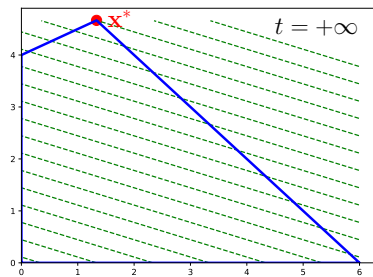
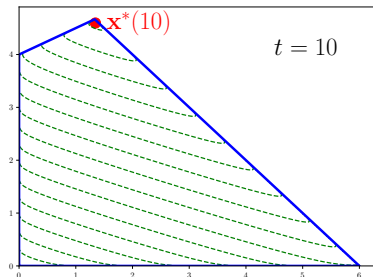
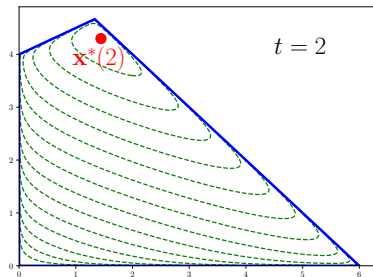
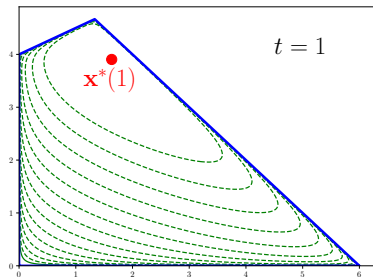


**Note.**  $\mathbf{x}^*(t) \in S$  is an interior point of the set defined by the inequality constraints.  $\mathbf{x}^*(t)$  is always a strictly feasible solution to (ICP).

**Note.** As  $t \rightarrow \infty$ ,  $\mathbf{x}^*(t)$  converges to the optimal solution  $\mathbf{x}^*$  of (ICP).

# Central point

Central points and level curves of  $f(x) + \frac{1}{t}B(x)$  for various values of  $t$ .



## Suboptimality gap of central point

**Lemma.** The solution  $\mathbf{x}^*(t)$  to (B) is  $\frac{m}{t}$ -suboptimal for (ICP), i.e.

$$f(\mathbf{x}^*(t)) - f^* \leq \frac{m}{t}$$

**Proof.** By the Lagrange condition for the problem (B), there exists  $\lambda^*$  s.t.

$$\nabla f(\mathbf{x}^*(t)) + \frac{1}{t} \nabla B(\mathbf{x}^*(t)) + \mathbf{A}^T \lambda^* = \mathbf{0}$$

Using the formula for  $\nabla B$  on slide 5, we obtain

$$\nabla f(\mathbf{x}^*(t)) + \sum_{j=1}^m \mu_j^* \nabla g_j(\mathbf{x}^*(t)) + \mathbf{A}^T \lambda^* = \mathbf{0} \quad (\dagger)$$

where  $\mu_j^* = -\frac{1}{tg_j(\mathbf{x}^*(t))}$ . Note  $\mu_j^* > 0$ , since  $\mathbf{x}^*(t)$  is strictly feasible.

Let  $\mathcal{L}(\mathbf{x}, \lambda, \mu)$  be the Lagrangian of (ICP). Then  $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*(t), \lambda^*, \mu^*) = \mathbf{0}$  by  $(\dagger)$ . Thus  $\mathbf{x}^*(t) = \operatorname{argmin}_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda^*, \mu^*)$ , since  $\mathcal{L}$  is convex in  $\mathbf{x}$ .

## Suboptimality of central point (cont'd)

For  $\mu \geq \mathbf{0}$  and  $\mathbf{x}$  in the feasible set  $X$ ,

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{j=1}^m \underbrace{\mu_j}_{\geq 0} \underbrace{g_j(\mathbf{x})}_{\leq 0} + \underbrace{(\mathbf{A}\mathbf{x} - \mathbf{b})^T \boldsymbol{\lambda}}_{=0} \leq f(\mathbf{x}) \quad (\ddagger)$$

In particular,

$$f(\mathbf{x}) \geq \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*), \quad \forall \mathbf{x} \in X$$

Minimizing over feasible  $\mathbf{x}$ ,

$$\begin{aligned} f^* &= \inf_{\mathbf{x} \in X} f(\mathbf{x}) \geq \inf_{\mathbf{x} \in X} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \quad \text{by } (\ddagger) \\ &\geq \mathcal{L}(\mathbf{x}^*(t), \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \quad \text{since } \mathbf{x}^*(t) \text{ minimizes } \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \\ &= f(\mathbf{x}^*(t)) + \sum_{j=1}^m \underbrace{\mu_j^* g_j(\mathbf{x}^*(t))}_{=-\frac{1}{t} \text{ by def. of } \mu_j^*} + \underbrace{(\mathbf{A}\mathbf{x}^*(t) - \mathbf{b})^T \boldsymbol{\lambda}^*}_{=0 \text{ by feasibility}} \\ &= f(\mathbf{x}^*(t)) - \frac{m}{t} \end{aligned}$$

# Interpretation via modified KKT conditions

$\mathbf{x}^*(t)$  satisfies the following modified KKT conditions for (ICP),

- feasibility

$$\mathbf{A}\mathbf{x}^*(t) = \mathbf{b}, \quad g_j(\mathbf{x}^*(t)) < 0, \quad j = 1, 2, \dots, m$$

- nonnegativity

$$\mu_j^* > 0, \quad j = 1, 2, \dots, m$$

- stationarity

$$\nabla f(\mathbf{x}^*(t)) + \mathbf{A}^T \boldsymbol{\lambda}^* + \sum_{j=1}^m \mu_j^* \nabla g_j(\mathbf{x}^*(t)) = \mathbf{0}$$

- approximate complementary slackness

$$\mu_j^* g_j(\mathbf{x}^*(t)) = -\frac{1}{t} \approx 0 \text{ for large } t, \quad j = 1, 2, \dots, m$$

## Barrier method

Since  $\mathbf{x}^*(t)$  is  $\frac{m}{t}$ -suboptimal, to achieve a desired accuracy, it suffices to solve

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) + \frac{\epsilon}{m} B(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \end{aligned}$$

However, this works well only for small problems, good initial points, and moderate accuracy ( $\epsilon$  not too small).

The **barrier method** or **path-following method** chooses an increasing sequence  $\{t_k\}$  and solves a sequence of problems, until  $t_k \geq m/\epsilon$ ,

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) + \frac{1}{t_k} B(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \end{aligned}$$

When solving for the  $k$ -th central point  $\mathbf{x}^*(t_k)$ , it uses the central point  $\mathbf{x}^*(t_{k-1})$  as the initial point.

## Barrier method (cont'd)

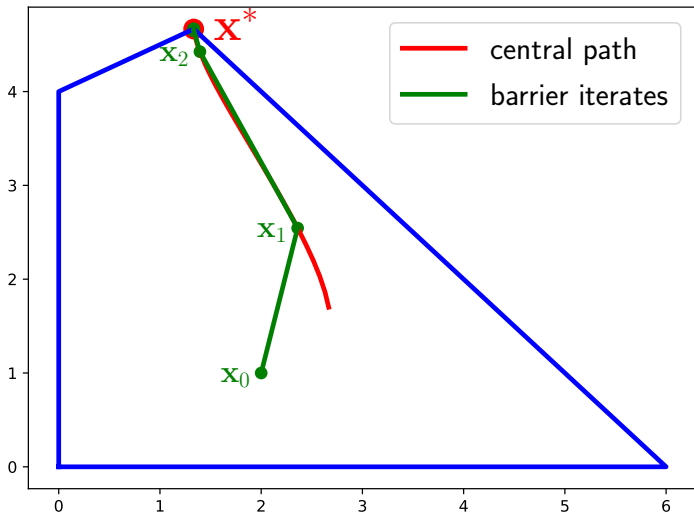
- 1: Choose a **strictly feasible**  $\mathbf{x}_0$  s.t.  $A\mathbf{x}_0 = \mathbf{b}$  and  $\mathbf{g}(\mathbf{x}_0) < \mathbf{0}$ , and  $t_0 > 0$
- 2: **for**  $k = 0, 1, 2, \dots$  **do**
- 3:      $\mathbf{x}_{k+1} \leftarrow \underset{\mathbf{x}: A\mathbf{x}=\mathbf{b}}{\operatorname{argmin}} \{f(\mathbf{x}) + \frac{1}{t_k}B(\mathbf{x})\}$  **starting from**  $\mathbf{x}_k$
- 4:     **if**  $t_k \geq m/\epsilon$  **then**
- 5:         **return**  $\mathbf{x}_{k+1}$
- 6:     **end if**
- 7:      $t_{k+1} \leftarrow \rho t_k$
- 8: **end for**

where  $\rho > 1$  and  $B$  is the log barrier function.

- On Line 3, we can solve the equality constrained problem by Newton's method or projected gradient descent. Newton's method is often recommended to better deal with ill-conditioning.
- The choice of  $\rho$  and  $t_0$  may affect how fast the algorithm converges. In practice, the performance is quite robust to the choice of  $\rho$  and  $t_0$ . [BV] suggests  $\rho \in [10, 20]$

## Example: LP

Solve the LP on slide 7 with  $t_0 = 0.3$ ,  $\mathbf{x}_0 = (2, 1)$ ,  $\rho = 10$ .



## Feasibility problem

How to find a strictly feasible initial point? By solving

$$\begin{aligned} \min_{\mathbf{x}, s} \quad & s \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\ & g_j(\mathbf{x}) \leq s, \quad j = 1, 2, \dots, m \end{aligned} \tag{F}$$

The original (ICP) is strictly feasible iff the optimal value  $s^*$  of (F) is negative.

How do we solve (F)? By the barrier method, since it is easy to find a strictly feasible point for (F).

- Pick an  $\mathbf{x}_0$  satisfying  $\mathbf{Ax}_0 = \mathbf{b}$  and then pick an  $s_0 > \max_{1 \leq j \leq m} g_j(\mathbf{x}_0)$ .

We do not need to solve (F) until optimality. Once we find a feasible  $(\mathbf{x}, s)$  with  $s < 0$ , we can stop and use  $\mathbf{x}$  as the initial point for (ICP).

## Feasibility problem (cont'd)

Alternatively, we can apply Newton's method with infeasible start to solve the following problem,

$$\begin{aligned} \min_{\mathbf{x}, s} \quad & f(\mathbf{x}) - \frac{1}{t} \sum_{j=1}^m \log(s - g_j(\mathbf{x})) \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b} \\ & s = 0 \end{aligned} \tag{F2}$$

The algorithm can be initialized with any  $\mathbf{x}_0 \in \text{dom} f \cap \bigcap_{j=1}^m \text{dom} g_j$  and  $s_0 > \max_{0 \leq j \leq m} g_j(\mathbf{x}_0)$ .

As long as the original problem is strictly feasible, Newton's method will eventually take a full Newton step with step size  $t = 1$ . After that,  $s = 0$  and  $\mathbf{x}$  will be feasible.

## Feasibility problem (cont'd)

If we don't know an  $\mathbf{x}_0 \in \text{dom} f \cap \bigcap_{j=1}^m \text{dom} g_j$  to start with, we can use a similar trick.

$$\begin{aligned} \min_{\mathbf{x}, s, \mathbf{z}} \quad & f(\mathbf{x} + \mathbf{z}_0) - \frac{1}{t} \sum_{j=1}^m \log(s - g_j(\mathbf{x} + \mathbf{z}_j)) \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\ & s = 0 \\ & \mathbf{z} = (z_0, \dots, z_m) = \mathbf{0} \end{aligned}$$

The drawback of this approach is that there is no good stopping criterion if the original problem is not feasible, but it becomes handy if we know feasibility a priori.